AFFINENESS OF THE STRUCTURE MORPHISMS IS EVENTUAL

THE STACKS PROJECT AUTHORS

Editorial note. Source typo kept verbatim: '(as a finite morphism is affine by definition)' where the argument concerns an affine morphism; recorded, not corrected in the counted span.

Allowed standard prerequisites (editorial).

- Limit affine detection. Limits, Lemma lemma-limit-affine (same chapter)

Source reference. Original lemma 01Z6: sources/stacks-chow/tags.txt, source lines 2581–2581. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01S8: sources/stacks-chow/tags.txt, source lines 2329–2329. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $X = \lim X_i$ be a directed limit of schemes over S with affine transition morphisms. Assume*

(1) *S quasi-compact and quasi-separated,*
(2) *X_i quasi-compact and quasi-separated,*
(3) *$X \to S$ affine.*

Then $X_i \to S$ is affine for i large enough.

Proof. Choose a finite affine open covering $S = \bigcup_{j=1,\ldots,n} V_j$. Denote $f : X \to S$ and $f_i : X_i \to S$ the structure morphisms. For each j the scheme $f^{-1}(V_j) = \lim_i f_i^{-1}(V_j)$ is affine (as a finite morphism is affine by definition). Hence by Lemma 01Z6 there exists an $i \in I$ such that each $f_i^{-1}(V_j)$ is affine. In other words, $f_i : X_i \to S$ is affine for i large enough, see Morphisms, Lemma 01S8. □

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.